# WORK BREAKDOWN STRUCTURE (WBS)

## MediConnect – Hospital Management System

*Document status: Initial planning draft*

## 1. Purpose

This Work Breakdown Structure (WBS) divides the MediConnect project into manageable phases and activities. It provides a basis for assigning responsibilities, estimating task durations, identifying dependencies and preparing the project Gantt chart.

## 2. Planning assumptions

- The WBS is a practice planning document based on the current MediConnect project scope.
- Task durations are preliminary working estimates, not approved commitments.
- The proposed project window is approximately 12 weeks. The detailed schedule must be checked against task dependencies, team availability, resource constraints and stakeholder approval.
- Some tasks may run in parallel where the required inputs, people and resources are available.
- The project scope and requirements must be validated and approved before the relevant design and development work is treated as authorised.

## 1.1 Project initiation

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.1.1 | Identify project stakeholders | Project Manager / Business Analyst | 1 day |
| 1.1.2 | Define project objectives | Project Manager / Business Analyst | 1 day |
| 1.1.3 | Prepare project charter | Project Manager | 2 days |
| 1.1.4 | Obtain project approval | Project Sponsor | 1 day |

## 1.2 Requirements gathering and analysis

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.2.1 | Conduct stakeholder interviews | Business Analyst | 3 days |
| 1.2.2 | Document business requirements | Business Analyst | 2 days |
| 1.2.3 | Analyse and prioritise requirements | Business Analyst | 2 days |
| 1.2.4 | Validate requirements with stakeholders | Business Analyst / Hospital Representatives | 2 days |
| 1.2.5 | Obtain requirements approval | Project Sponsor / Relevant Stakeholders | 1 day |

## 1.3 System design

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.3.1 | Prepare current and proposed process flows | Business Analyst | 2 days |
| 1.3.2 | Design user interface and wireframes | UI/UX Designer | 5 days |
| 1.3.3 | Design database structure | Database Developer | 5 days |
| 1.3.4 | Design system architecture and security controls | Technical Lead / Security Team | 5 days |
| 1.3.5 | Review and approve system designs | Project Manager / Business Analyst / Technical Lead | 2 days |

## 1.4 System development

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.4.1 | Set up development environment | Development Team | 3 day |
| 1.4.2 | Develop patient registration module | Developers | 5 days |
| 1.4.3 | Develop appointment management module | Developers | 5 days |
| 1.4.4 | Develop medical records module | Developers | 6 days |
| 1.4.5 | Develop billing and payment module | Developers | 5 days |
| 1.4.6 | Implement user roles and access controls | Developers / Security Team | 5 days |
| 1.4.7 | Develop management dashboard and reports | Developers | 14 days |
| 1.4.8 | Integrate system modules | Developers / Technical Lead | 14 days |

## 1.5 Testing and quality assurance

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.5.1 | Prepare test cases and test data | QA Tester | 2 days |
| 1.5.2 | Conduct functional testing | QA Tester | 4 days |
| 1.5.3 | Conduct security testing | Security Team / QA Tester | 3 days |
| 1.5.4 | Conduct integration testing | QA Tester / Developers | 5 days |
| 1.5.5 | Fix identified defects | Developers | 4 days |
| 1.5.6 | Perform regression testing | QA Tester | 5 days |

## 1.6 User acceptance testing (UAT)

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.6.1 | Prepare UAT scenarios | Business Analyst / QA Tester | 2 days |
| 1.6.2 | Conduct UAT with hospital staff | Hospital Representatives | 7 days |
| 1.6.3 | Document feedback and outstanding issues | Business Analyst / QA Tester | 3 days |
| 1.6.4 | Obtain UAT approval | Hospital Management / Project Sponsor | 1 day |

## 1.7 Deployment

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.7.1 | Prepare production environment | Technical Team | 2 days |
| 1.7.2 | Migrate approved patient data | Database / Technical Team | 3 days |
| 1.7.3 | Deploy the system | Developers / Technical Lead | 1 day |
| 1.7.4 | Verify production functionality | QA Tester / Technical Team | 7 days |

## 1.8 Training and handover

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.8.1 | Prepare user documentation and guides | Business Analyst / Technical Writer | 7 days |
| 1.8.2 | Train hospital staff | Training Team / Project Team | 7 days |
| 1.8.3 | Hand over system and technical documentation | Technical Lead | 1 day |

## 1.9 Project closure

| WBS ID | Activity / Task | Responsible role | Estimated duration |
|---|---|---|---|
| 1.9.1 | Obtain final project acceptance | Project Manager / Project Sponsor | 1 day |
| 1.9.2 | Document lessons learned | Project Manager / Project Team | 5 day |
| 1.9.3 | Complete project closure report | Project Manager | 2 day |

## 3. Key dependencies to consider

- Requirements validation and approval should be completed before the team finalises the system design.
- The relevant designs and technical decisions should be ready before development of the related modules begins.
- System integration depends on the availability of the modules and interfaces being integrated.
- Functional, security and integration testing depend on the relevant development work being sufficiently complete.
- User acceptance testing should follow system testing and resolution of critical defects.
- Deployment should take place only after the required UAT approval and deployment-readiness checks.
- Staff training and handover should be coordinated with deployment readiness and the availability of approved user documentation.

## 4. Review and approval

This document is an initial planning draft for the MediConnect practice project. It has not been approved by a real hospital or project sponsor. The WBS, estimates, responsibilities and dependencies should be reviewed and updated before being used as a committed project baseline.